

Lecture 26: MAP Estimator

Jamie Haddock

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- 1 Bayesian estimation::: {.callout-note icon=false}### Definition: Prior, posterior, MAPThe parameter θ has a prior $f(\theta)$, updated via Bayes' rule (L8) into a posterior $f(\theta | \text{data}) \propto L(\text{data} | \theta)f(\theta)$. The MAP estimator is

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} [\log L(\text{data} | \theta) + \log f(\theta)].$$

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- 1.1 Reading & HomeworkReading: §8 (Bertsekas & Tsitsiklis, *Introduction to Probability*).Due today: Homework 12 (Gradescope). 1

—title: "Lecture 26: MAP Estimator"author: "Jamie Haddock"format: revealjs: output-file: Lecture26_slides slide-number: true preview-links: auto logo: figs/hmc.png css: input/slides.css incremental: true smaller: true code-fold: true embed-resources: true html: code-fold: true embed-resources: true pdf: documentclass: article toc: true number-sections: true geometry: - top=1in - left=1in - bottom=1in - right=1informat-links: falsejupyter: python3filters: - input/remove-pause.luaexecute: echo: true eval: true—

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```
from scipy.stats import beta as beta_disttheta_true = 0.65prior_a, prior_b = 2, 2fig, axes = plt.subplots
```

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